EXERCISE 3

1.

```
SELECT product_name,
        price,
CASE
   WHEN price < 100 THEN 'Budget'
   WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
ELSE 'Expensive'
END AS price_category
FROM products;
```

product_name	price	price_category
laptop	1200	expensive
phone	800	mid-range
keyboard	45	Budget
monitor	300	Mid-range
mouse	25	Budget

2.

```
SELECT customer_name,
        amount,
CASE
WHEN amount < 500 THEN 'Low value'
WHEN amount ≥ 500 AND amount < 1000 THEN 'Medium value'
ELSE 'High value'
END AS order_value category
FROM orders;
```

Orders

customer_name	amount	order_value_category
Alice	150	Medium value
Bob	560	Medium value
Charlie	999.99	High value
Diana	45	Low value
Ethan	1200	High value

3.

```
SELECT emp_name,
       department,
       salary,
CASE
WHEN department = 'IT' AND salary > 8000 THEN 'SENIOR IT'
WHEN department = 'HR' AND salary > 5000 THEN 'Experienced HR'
ELSE 'Staff'
END AS position_level
FROM employees;
```

employees

emp_name	department	salary	position_level
John	IT	85000	SENIOR IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4.

```
SELECT student_name,
        score,
CASE
WHEN score ≥ 90 THEN 'A'
WHEN score BETWEEN 80 AND 89 THEN 'B'
WHEN score BETWEEN 70 AND 79 THEN 'C'
WHEN score BETWEEN 60 AND 69 THEN 'D'
ELSE 'F'
END AS grade
FROM students;
```

students

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5.

```
SELECT delivery_id,
        delivery_time_minutes,
CASE
WHEN delivery_time_minutes ≤ 30 THEN 'Fast'
WHEN delivery_time_minutes 31 AND 60 THEN 'on TIME'
ELSE 'late'
END AS Performance
FROM deliveries;
```

deliveries

delivery_id	delivery_time_minutes	Performance
1	45	late
2	80	late
3	30	fast
4	65	late
5	100	late

6.

```
SELECT issue_type,
       Priority,
CASE
WHEN Priority = 3 THEN 'High'
WHEN Priority = 2 THEN 'Medium'
WHEN Priority = 1 THEN 'Low'
ELSE unknown
END AS Priority_label
FROM tickets;
```

tickets

issue_type	Priority	Priority_label
login issue	1	low
Server down	3	high
Slow system	2	medium
Email error	2	medium
Password reset	1	low

7.

```
SELECT student_id,
        (days_present/total_days)X100 AS attendance_percentage
CASE
WHEN attendance_percentage ≥ 90 THEN 'Excellent'
WHEN attendance_percentage BETWEEN 75 AND 89 THEN 'Good'
ELSE 'Needs Improvement'
END AS attendance_status
FROM attendance
GROUP BY student_id,
        days_present
        total_days
LIMIT 10;
```

a attendance

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Need Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

8.

```
SELECT product_id,
        stock_qty,
CASE
WHEN stock_qty = 0 THEN 'Out of Stock'
WHEN stock qty BETWEEN 1 AND 5 THEN 'Low stock'
ELSE 'in stock'
END AS stock_status
FROM products_inventory;
```

Product_inventory

Product_id	stock_qty	Stock_status
1	5	Low stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

9.

```
SELECT subject,
        enrolled_students,
CASE
WHEN enrolled_students ≥ 25 THEN 'large'
WHEN enrolled_students BETWEEN 10 AND 25 THEN 'medium'
ELSE 'small'
END AS class_size_category
FROM classes;
```

Classes

subject	enrolled_students	class_size_category
math	30	large
English	25	large
Science	15	medium
Art	5	small
History	20	medium

10

```
SELECT payment_id,
        payment_method,
        amount,
CASE
WHEN payment_method='cash' AND amount ≥ 200
THEN 'Eligible for Discount'
ELSE 'Not Eligible'
END AS discount_Eligibility
FROM payments;
```

Payments

Payment_id	Payment_method	amount	discount_Eligibility
1	Card	50	Not Eligible
2	Cash	200	Eligible for discount
3	Card	150	Not Eligible
4	PayPal	75	Not Eligible
5	Cash	300	Eligible for discount